

Andrew Michael Collins

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Staff Software Engineer

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Full Stack Engineer & AI Specialist with over 7 years of experience building robust, scalable applications using the MERN stack and Python backend like FastAPI and Django. Skilled in integrating AI technologies like fine-tuned LLM models and RAG pipelines to improve system performance and user experience. Experienced with cloud platforms (AWS, Azure, GCP) and MLOps, ensuring smooth deployment of AI models. With expertise in Ruby, Go, Java, and .NET/C#, I lead development projects from start to finish, always focusing on delivering high-quality solutions using Agile and SAFe practices.

Experience

Nationwide Marketing Group, NC, United States

Staff Software engineer

3/2020 – June 2025

Led the development and optimization of full-stack applications, specializing in the **MERN stack and Python backend**, while integrating **AI-driven features**, building **cloud infrastructure**, and applying **DevOps practices** to create scalable, high-performance solutions.

- Built and optimized full-stack applications with the MERN stack (MongoDB, Express, React, Node.js) and Python, improving system performance and reducing load times by 20%, which led to a smoother, faster experience for users.
- Developed backend services using Python frameworks like FastAPI, Flask, and Django, streamlining data processing and boosting overall performance by 25%, making the system more responsive and better equipped for future growth.
- Integrated AI-powered features, including recommendation systems and chatbots, which increased customer engagement by 15%, automating key tasks and making the user experience more personalized.
- Designed and deployed cloud infrastructure on AWS and Azure, automating the provisioning process and cutting deployment times by 40%, helping the system scale quickly to meet business demands.
- Built and maintained RESTful APIs, ensuring smooth communication and integration across platforms, improving the reliability and flexibility of the entire system.
- Enhanced AI-driven search functionality by fine-tuning models, improving search accuracy and relevance by 25%, making it easier for users to find exactly what they need faster.
- Led DevOps initiatives using Terraform and Ansible, automating infrastructure management to ensure deployments were consistent, reliable, and repeatable.
- Built and optimized CI/CD pipelines, speeding up software delivery and ensuring high-quality code through automated testing, reducing the time it took to release new features.
- Collaborated closely with cross-functional teams, integrating AI, cloud, and DevOps solutions to deliver projects that were aligned with business goals and delivered on time.
- Skilled in Go, Java, and C#, using these languages to optimize backend performance, ensuring systems remained fast, scalable, and reliable as the business grew.

Education

Bachelor Degree in Computer Science, Baylor University
Waco, TX, United States (2016 - 2020)

Skills

Programming Languages: Python, JavaScript/TypeScript, Go, Java, C#, SQL

Full-Stack & Backend Development:

- **Backend:** FastAPI, Flask, Django, Node.js, Express, Spring Boot, ASP.NET, Microservices
- **Frontend:** React, React Native, Next.js, Angular, Vue.js, Micro-Frontend Architecture
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB, Firebase, DynamoDB, Redis, Neo4j
- **API Development & Integration:** RESTful APIs, GraphQL, WebSocket, API & Webhook Integration

AI/ML & Data Science:

- LLMs, RAG, NLP, NLU, Hugging Face, LangChain, PyTorch, TensorFlow, Keras, LSTM, Model Training, GPU Optimization, Multi-modal AI, Predictive Analytics

DevOps & Cloud:

- **Cloud Platforms:** AWS, Azure, GCP
- **DevOps Tools:** Docker, Kubernetes, Terraform, Ansible, Jenkins, CI/CD, Nginx
- **Infrastructure Automation:** Terraform, Ansible
- **Version Control:** Git, GitHub, GitLab

Automation & Tools:

- ServiceNow, Microsoft Teams Bot, Flowise, API & Webhook Integration
- Monitoring & Dashboards:
- Grafana, Tableau, Logging/Alerting Frameworks, Metrics & Observability